THIN AND THICK CYLINDERS

In many engineering applications, cylinders are frequently used for transporting or storing of liquids, gases or fluids.

Eg: Pipes, Boilers, storage tanks etc.

- A cylinder or spherical shell is considered to be thin when the metal thickness is small compared to internal diameter.

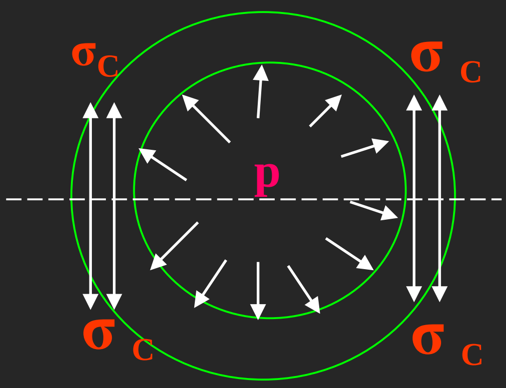
- i. e., when the wall thickness, ‘t’ is equal to or less than ‘d/20’, where ‘d’ is the internal diameter of the cylinder or shell, we consider the cylinder or shell to be thin, otherwise thick.

When a cylinder is subjected to a internal pressure, at any point on the cylinder wall, three types of stresses are induced on three mutually perpendicular planes.

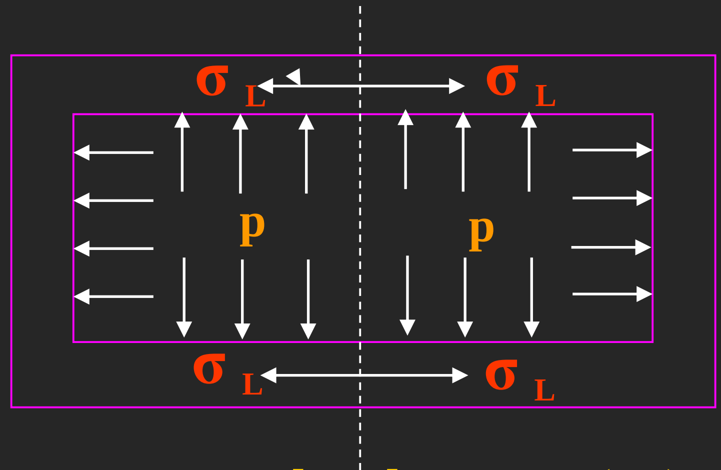
1. **Hoop or Circumferential Stress (σ_C)** – This is directed along the tangent to the circumference and tensile in nature. Thus, there will be increase in diameter.

2. **Longitudinal Stress (σ_L)** – This stress is directed along the length of the cylinder. This is also tensile in nature and tends to increase the length.

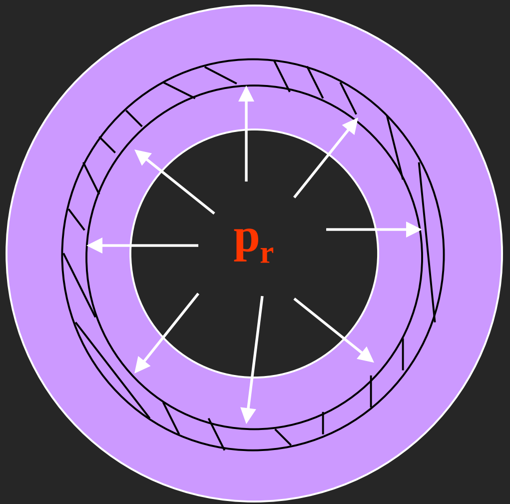
3. **Radial pressure (p_r)** – It is compressive in nature. Its magnitude is equal to fluid pressure on the inside wall and zero on the outer wall if it is open to atmosphere.



1. Hoop Stress (♦$_{C}$)

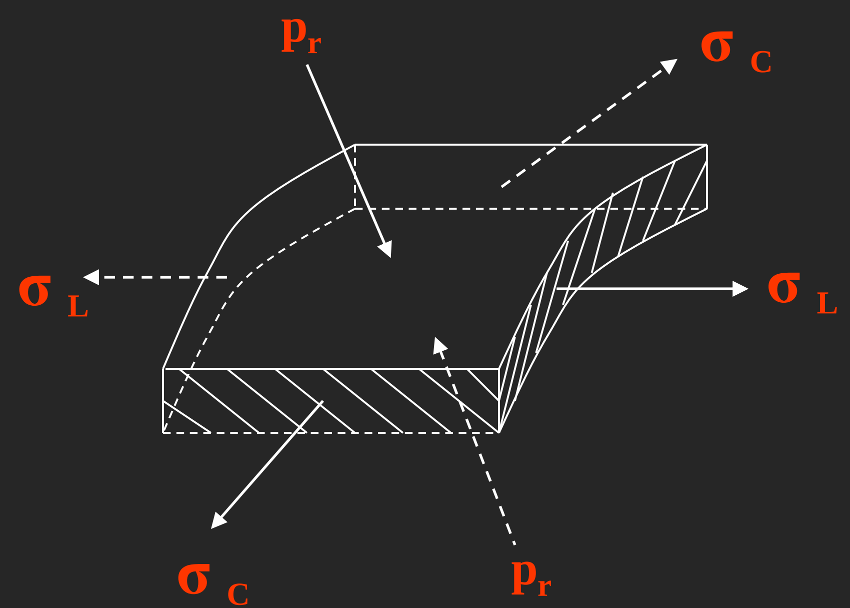


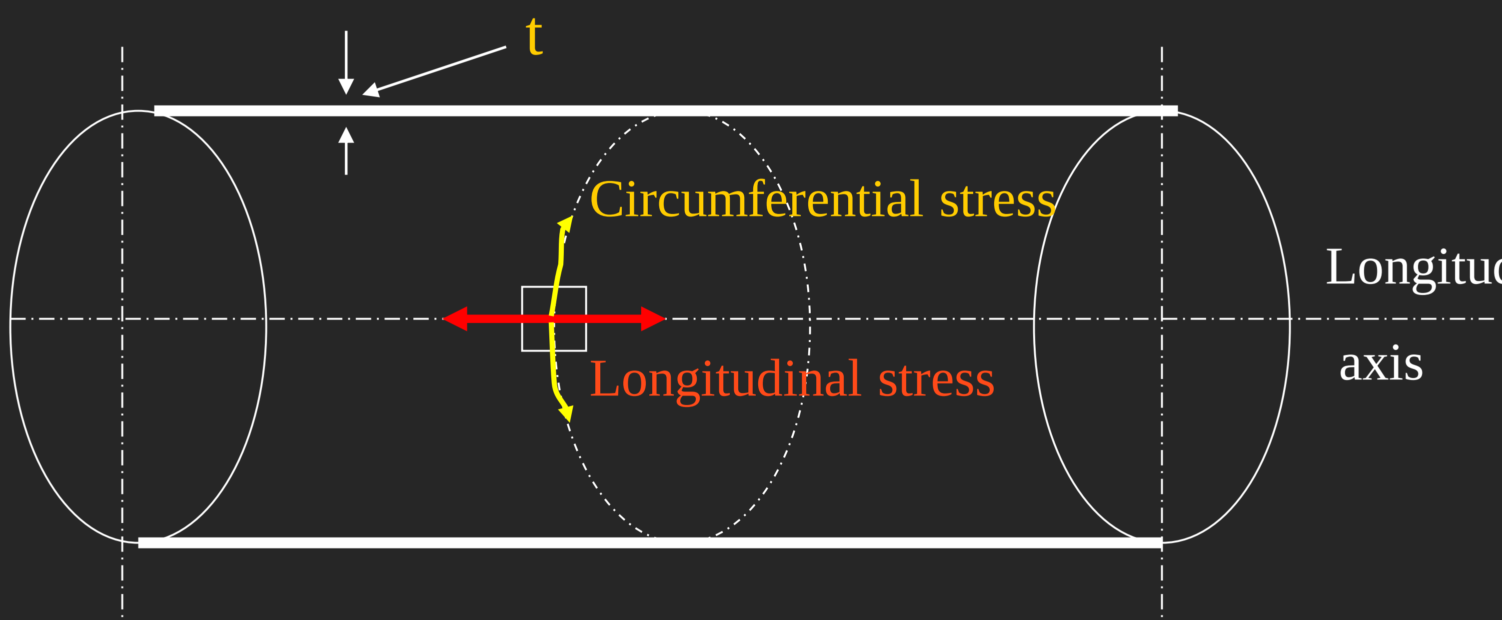
2. Longitudinal Stress (♦$_{L}$)



3. Radial Stress (p_r)

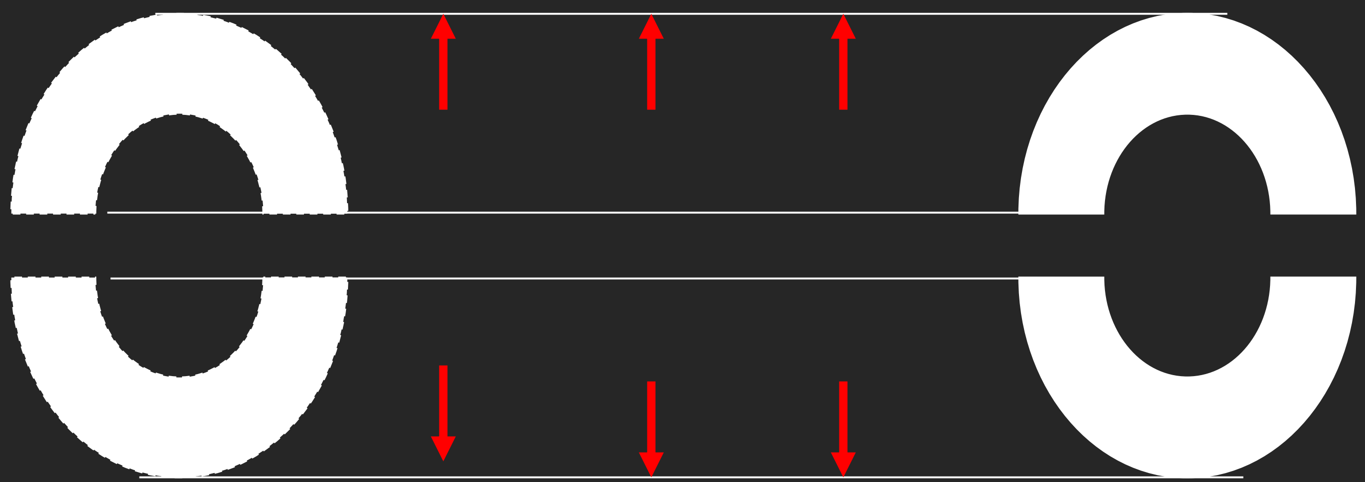
Element on the cylinder wall subjected to these three stresses

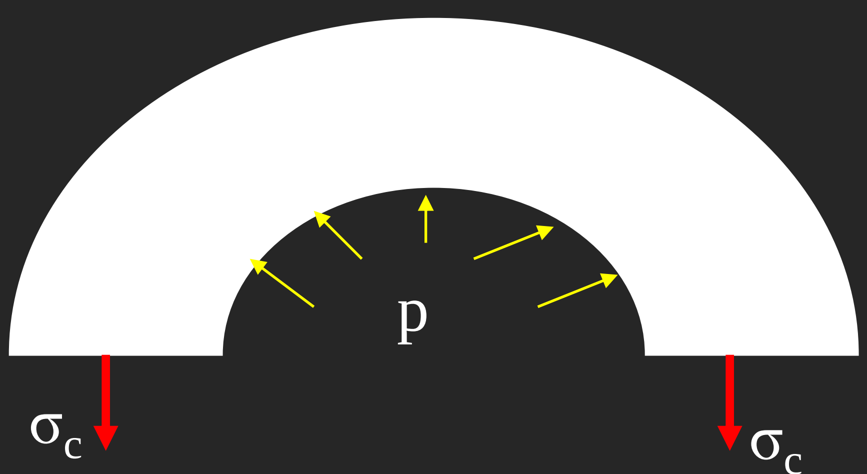




The stress acting along the circumference of the cylinder is called circumferential stresses whereas the stress acting along the length of the cylinder (i.e., in the longitudinal direction) is known as longitudinal stress

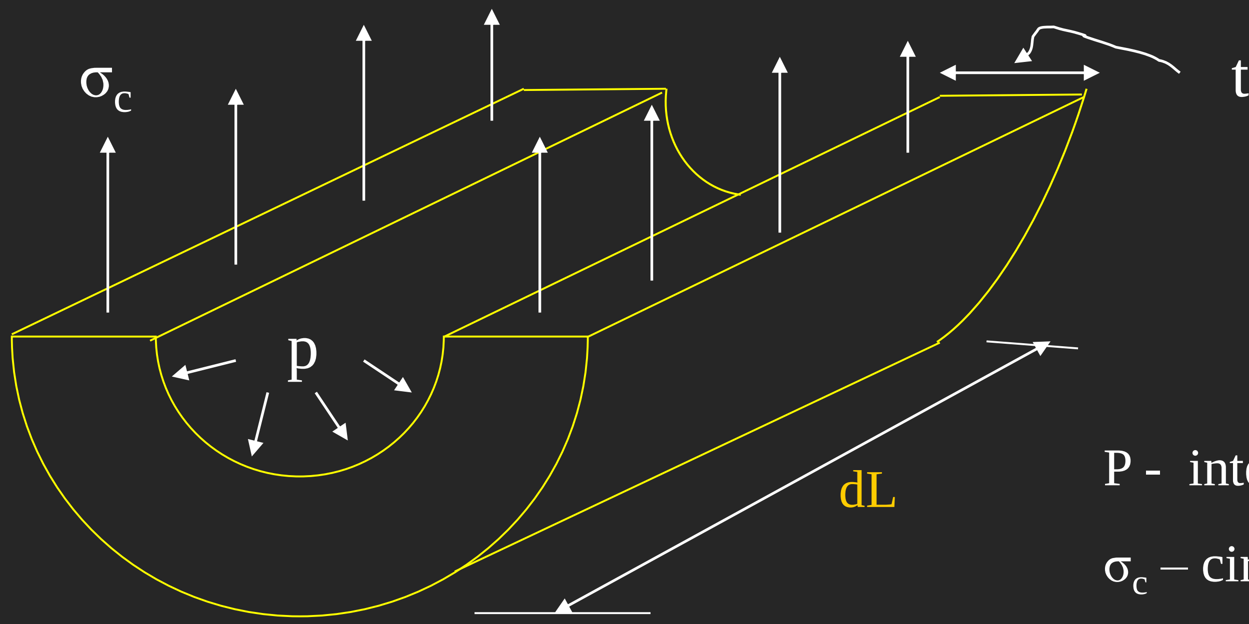
The bursting will take place if the force due to internal (fluid) pressure (acting vertically upwards and downwards) is more than the resisting force due to circumferential stress set up in the material.





P - internal pressure (stress)

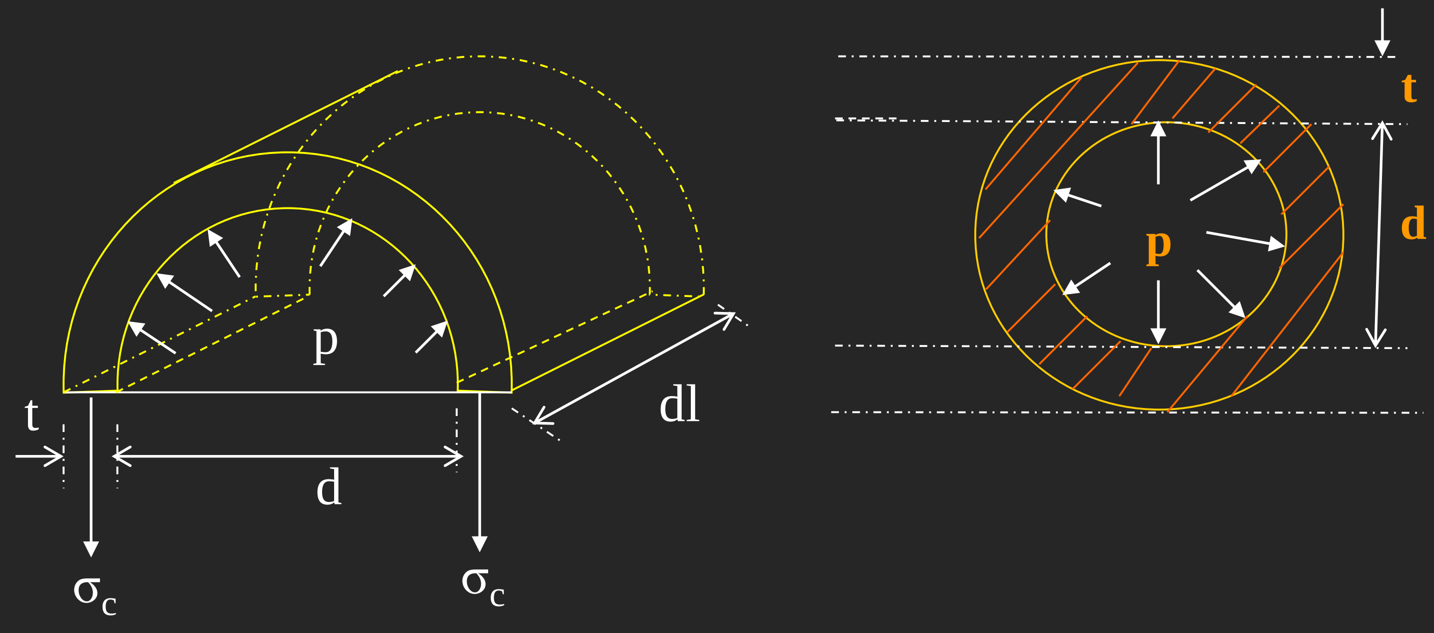
σ_c –circumferential stress



P - internal pressure (stress)

σ_c – circumferential stress

EVALUATION OF CIRCUMFERENTIAL or HOOP STRESS (σ_C):



Consider a thin cylinder closed at both ends and subjected to internal pressure 'p' as shown in the figure.

Let d=Internal diameter, t = Thickness of the wall

L = Length of the cylinder.

<u>To determine the Bursting force across the diameter:</u>

Force due to fluid pressure = p × area on which p is acting

= p ×(d ×L) Taking projected area

(bursting force)

Force due to circumferential stress = σ_c × area on which σ_c is acting

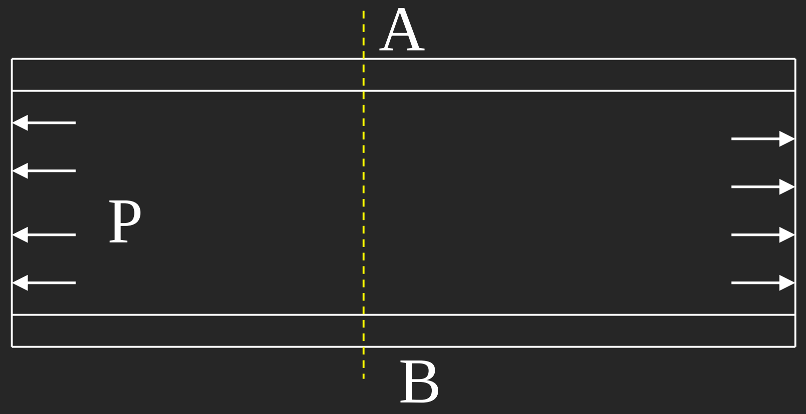
resisting force) = $\sigma_c \times (L \times t + L \times t) = \sigma_c \times 2 L \times t$

Under equilibrium bursting force = resisting force

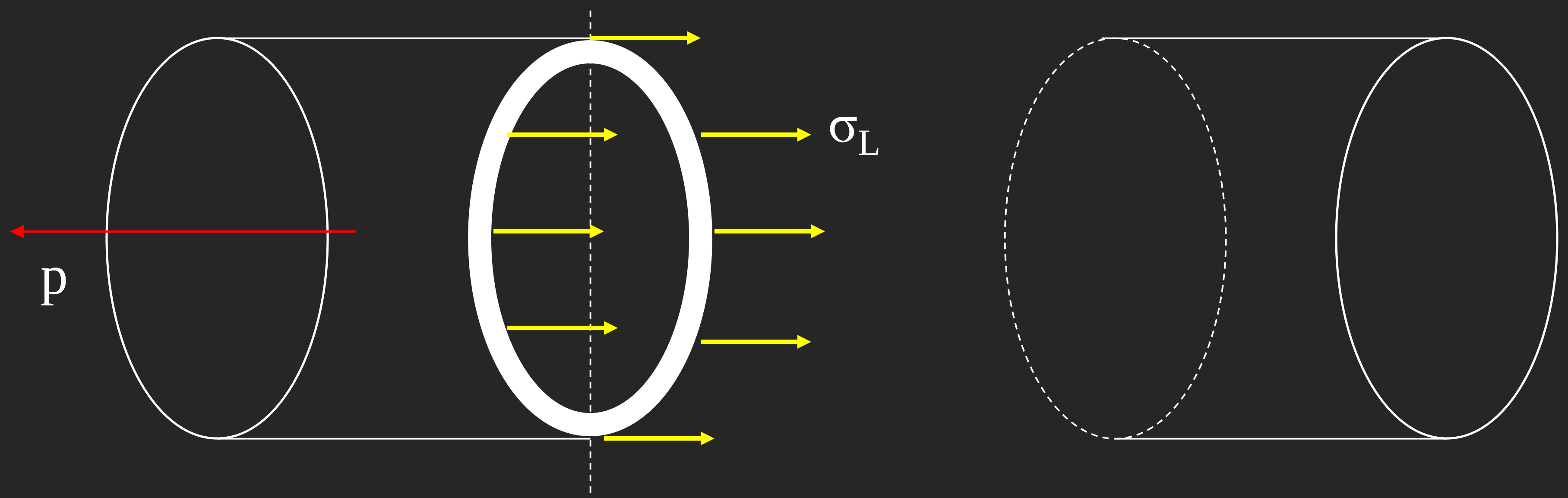
$$p \times (d \times L) = \sigma_c \times 2 L \times t$$

$$\therefore \text{Circumferential stress, } \sigma_c = \frac{p \times d}{2 \times t} \quad \text{..........................(1)}$$

LONGITUDINAL STRESS (σ_L):

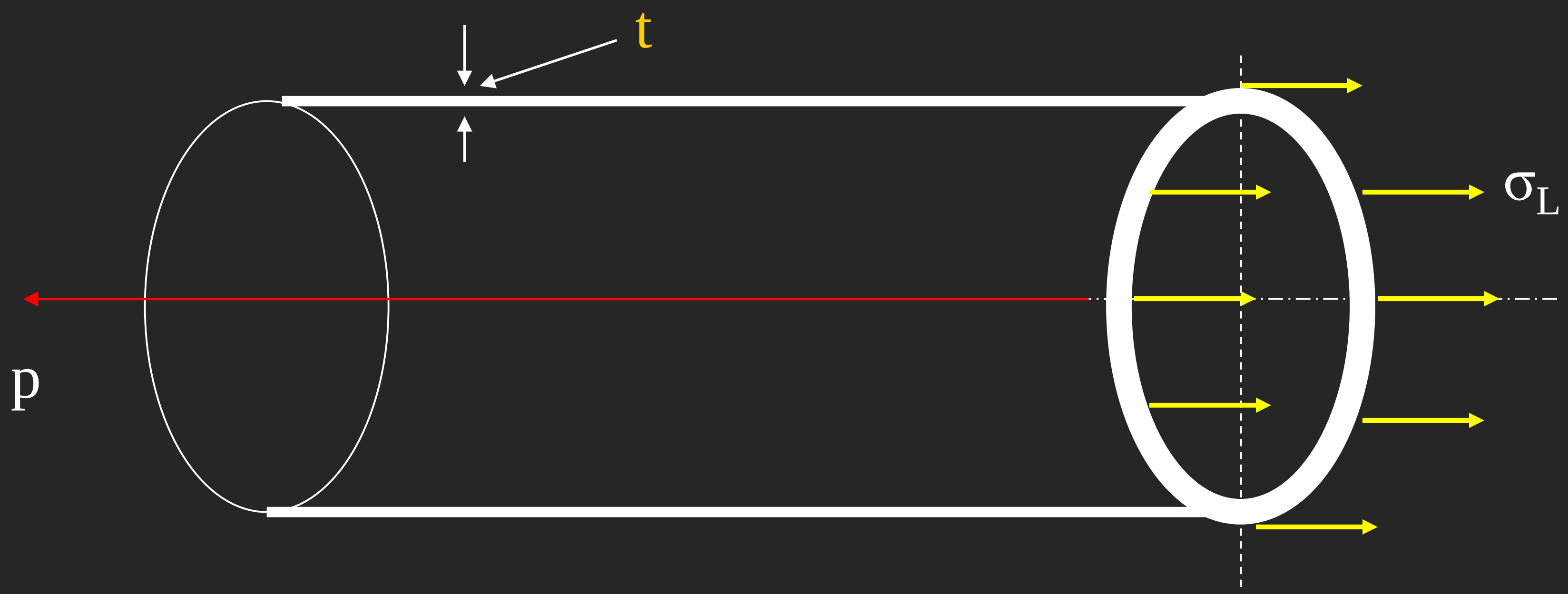


The bursting of the cylinder takes place along the section AB



The force, due to pressure of the fluid, acting at the ends of the thin cylinder, tends to burst the cylinder as shown in figure

EVALUATION OF LONGITUDINAL STRESS (σ_L):



$$\text{Longitudinal bursting force (on the end of cylinder)} = p \times \frac{\pi}{4} \times d^2$$

$$\text{Area of cross section resisting this force} = \pi \times d \times t$$

$$\text{Let } \sigma_L = \text{Longitudinal stress of the material of the cylinder.}$$

$$\therefore \text{Resisting force} = \sigma_L \times \pi \times d \times t$$

Under equillibrium, bursting force = resisting force

$$\text{i.e., } p \times \frac{\pi}{4} \times d^2 = \sigma_L \times \pi \times d \times t$$

$$\boxed{\therefore \quad \text{Longitudinal stress, } \sigma_L = \frac{p \times d}{4 \times t}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

$$\boxed{\text{From eqs (1) \& (2), } \quad \underline{\underline{\sigma_C = 2 \times \sigma_L}}}$$

$$\text{Force due to fluid pressure} = p \times \text{area on which p is acting}$$

$$= p \times \frac{\pi}{4} \times d^2$$

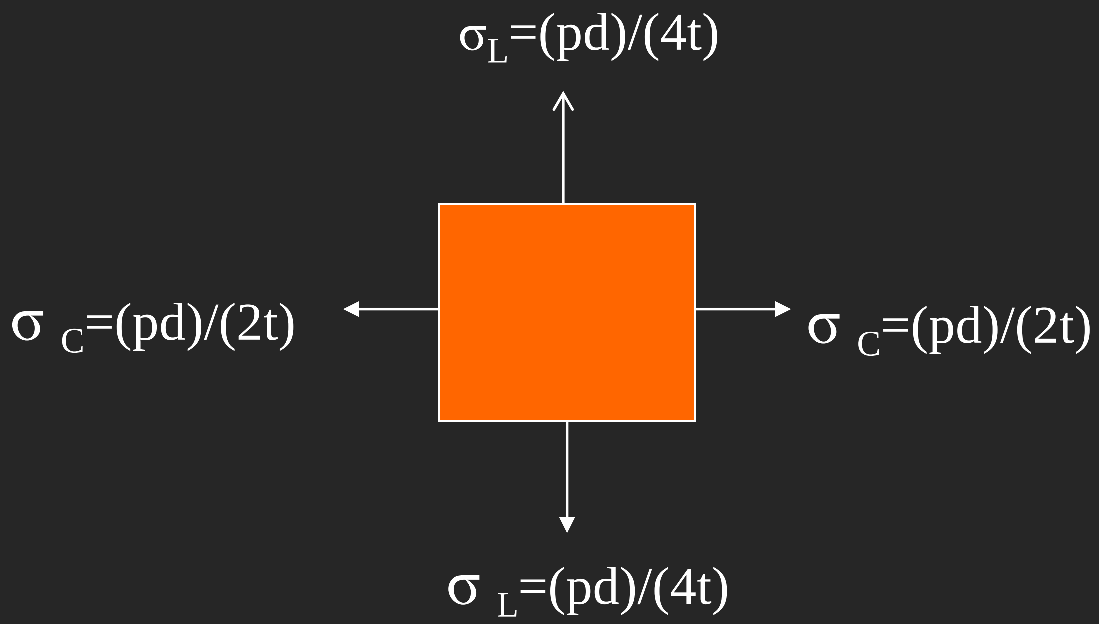
$$\text{Re}\,\textit{sisting force} = \sigma_L \times \textit{area on which } \sigma_L \textit{ is acting}$$

$$= \sigma_L \times \underbrace{\boxed{\pi \times d}}_{\text{circumference}} \times t$$

$$\text{Under equillibrium, bursting force} = \text{resisting force}$$

$$\therefore \quad \text{Longitudinal stress, } \sigma_L = \frac{p \times d}{4 \times t} \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

EVALUATION OF STRAINS



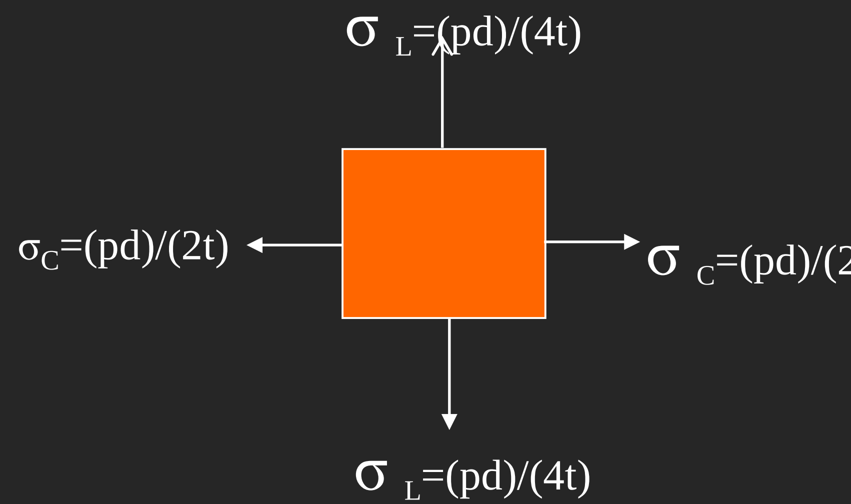
A point on the surface of thin cylinder is subjected to biaxial stress system, (Hoop stress and Longitudinal stress) mutually perpendicular to each other, as shown in the figure. The strains due to these stresses i.e., circumferential and longitudinal are obtained by applying Hooke’s law and Poisson’s theory for elastic materials.

Circumferential strain, ε_C :

$$\varepsilon_C = \frac{\sigma_C}{E} - \mu \times \frac{\sigma_L}{E}$$

$$= 2 \times \frac{\sigma_L}{E} - \mu \times \frac{\sigma_L}{E}$$

$$= \frac{\sigma_L}{E} \times (2 - \mu)$$



i.e., $$\varepsilon_C = \frac{\delta d}{d} = \frac{p \times d}{4 \times t \times E} \times (2 - \mu) \qquad \ldots\ldots(3)$$

Note: Let δd be the change in diameter. Then

$$\varepsilon_c = \frac{final\ circumference - original\ circumference}{original\ circumference}$$

$$= \left[\frac{\pi(d + \delta d) - \pi d}{\pi d}\right] = \frac{\delta d}{d}$$

Longitudin al strain, ε_L :

$$\varepsilon_L = \frac{\sigma_L}{E} - \mu \times \frac{\sigma_C}{E}$$

$$= \frac{\sigma_L}{E} - \mu \times \frac{(2 \times \sigma_L)}{E} = \frac{\sigma_L}{E} \times (1 - 2 \times \mu)$$

$$\boxed{\text{i.e.,} \quad \varepsilon_L = \frac{\delta l}{L} = \frac{p \times d}{4 \times t \times E} \times (1 - 2 \times \mu) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(4)}$$

VOLUMETRIC STRAIN, $\frac{\delta V}{V}$

Change in volume = δV = final volume – original volume

original volume = V = area of cylindrical shell × length

$$= \frac{\pi d^2}{4} L$$

final volume = final area of cross section × final length

$$= \frac{\pi}{4}\left[d + \delta d\right]^2 \times \left[L + \delta L\right]$$

$$= \frac{\pi}{4}\left[d^2 + (\delta d)^2 + 2d\,\delta d\right] \times \left[L + \delta L\right]$$

$$= \frac{\pi}{4}\left[d^2 L + (\delta d)^2 L + 2L\,d\,\delta d + d^2\,\delta L + (\delta d)^2\,\delta L + 2d\,\delta d\;\delta L\right]$$

neglecting the smaller quantities such as $(\delta d)^2 L, (\delta d)^2\,\delta L$ *and* $2d\,\delta d\;\delta L$

$$\text{Final volume} = \frac{\pi}{4}\left[d^2 L + 2L\,d\,\delta d + d^2\delta L\right]$$

$$\text{change in volume } \delta V = \frac{\pi}{4}\left[d^2 L + 2L\,d\,\delta d + d^2\delta L\right] - \frac{\pi}{4}\left[d\right]^2 L$$

$$\delta V = \frac{\pi}{4}\left[2L\,d\,\delta d + d^2\delta L\right]$$

$$\frac{dv}{V} = \frac{\frac{\pi}{4}\left[2\,d\,L\,\delta\,d + \delta\,L\,d^{2}\right]}{\frac{\pi}{4} \times d^{2} \times L}$$

$$= \frac{\delta L}{L} + 2 \times \frac{\delta d}{d}$$

$$\frac{\mathbf{dV}}{\mathbf{V}} = \varepsilon_L + 2 \times \varepsilon_C$$

$$= \frac{p \times d}{4 \times t \times E}(1 - 2 \times \mu) + 2 \times \frac{p \times d}{4 \times t \times E}(2 - \mu)$$

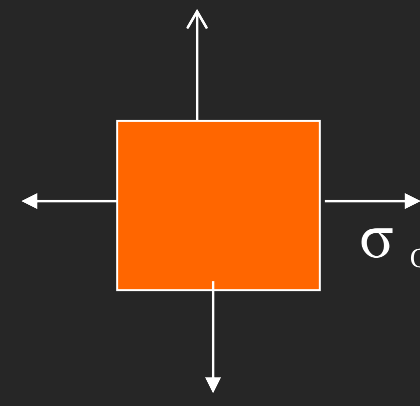
$$\text{i.e.,} \quad \boxed{\underline{\underline{\frac{dv}{V} = \frac{p \times d}{4 \times t \times E}(5 - 4 \times \mu)}}\ldots\ldots\ldots\ldots\ldots\ldots(5)}$$

Maximum Shear stress :

There are two principal stresses at any point, viz., Circumferential and longitudinal. Both these stresses are normal and act perpendicular to each other.

$$\therefore \text{ Maximum Shear stress, } \tau_{max} = \frac{\sigma_C - \sigma_L}{2}$$

$$= \frac{\frac{pd}{2t} - \frac{pd}{4t}}{2}$$



i.e., $\underline{\underline{\tau_{max} = \frac{pd}{8t}}}$ ....(5)

ILLUSTRATIVE PROBLEMS

PROBLEM 1:

A thin cylindrical shell is 3m long and 1m in internal diameter. It is subjected to internal pressure of 1.2 MPa. If the thickness of the sheet is 12mm, find the circumferential stress, longitudinal stress, changes in diameter, length and volume . Take E=200 GPa and μ= 0.3.

SOLUTION:

1. Circumferential stress, σ_C:

$$\sigma_C = (p\times d) / (2\times t)$$
$$= (1.2\times 1000) / (2\times 12)$$
$$= \underline{50\ N/mm^2 = 50\ MPa}\ \text{(Tensile).}$$

2. Longitudinal stress, σ_L:

$$\sigma_L = (p\times d) / (4\times t)$$
$$= \sigma_C/2 = 50/2$$
$$= \underline{25\ N/mm^2 = 25\ MPa}\ \text{(Tensile).}$$

3. Circumferential strain, ε_c:

$$\varepsilon_c = \frac{(p \times d)}{(4 \times t)} \times \frac{(2-\mu)}{E}$$

$$= \frac{(1.2 \times 1000)}{(4 \times 12)} \times \frac{(2-0.3)}{200 \times 10^3}$$

$$= \underline{2.125 \times 10^{-04}} \text{ (Increase)}$$

Change in diameter, $\delta d = \varepsilon_c \times d$

$= 2.125 \times 10^{-04} \times 1000 = \underline{0.2125}$ mm (Increase).

4. Longitudinal strain, ε_L:

$$\varepsilon_L = \frac{(p \times d)}{(4 \times t)} \times \frac{(1-2 \times \mu)}{E}$$

$$= \frac{(1.2 \times 1000)}{(4 \times 12)} \times \frac{(1-2 \times 0.3)}{200 \times 10^3}$$

$$= \underline{5 \times 10^{-05}} \text{ (Increase)}$$

Change in length $= \varepsilon_L \times L = 5 \times 10^{-05} \times 3000 = \underline{0.15}$ mm (Increase).

Volumetric strain, $\frac{dv}{V}$:

$$\frac{dv}{V} = \frac{(p \times d)}{(4 \times t) \times E} \times (5 - 4 \times \mu)$$

$$= \frac{(1.2 \times 1000)}{(4 \times 12) \times 200 \times 10^{3}} \times (5 - 4 \times 0.3)$$

$$= 4.75 \times 10^{-4} \text{ (Increase)}$$

$$\therefore \text{Change in volume,} \quad dv = 4.75 \times 10^{-4} \times V$$

$$= 4.75 \times 10^{-4} \times \frac{\pi}{4} \times 1000^{2} \times 3000$$

$$= 1.11919 \times 10^{6} \text{ mm}^{3} = 1.11919 \times 10^{-3} \text{ m}^{3}$$

$$= \underline{1.11919 \text{ Litres}}.$$

A copper tube having 45mm internal diameter and 1.5mm wall thickness is closed at its ends by plugs which are at 450mm apart. The tube is subjected to internal pressure of 3 MPa and at the same time pulled in axial direction with a force of 3 kN. Compute: i) the change in length between the plugs ii) the change in internal diameter of the tube. Take E_{CU} = 100 GPa, and μ_{CU} = 0.3.

SOLUTION:

A] Due to Fluid pressure of 3 MPa:

Longitudinal stress, $\sigma_L = (p\times d) / (4\times t)$

$= (3\times 45) / (4\times 1.5) = 22.50 \text{ N/mm}^2 = 22.50 \text{ MPa}.$

$$\text{Long. strain, } \varepsilon_L = \frac{(p\times d)}{4\times t}\times\frac{(1-2\times\mu)}{E}$$

$$= \frac{22.5\times(1-2\times 0.3)}{100\times 10^3} = \underline{9\times 10^{-5}}$$

Change in length, $\delta_L = \varepsilon_L \times L = 9\times 10^{-5}\times 450$ = <u>+0.0405 mm</u> (increase)

$\boxed{Pd/4t = 22.5}$

$$\text{Circumferential strain } \varepsilon_C = \frac{(p \times d)}{(4 \times t)} \times \frac{(2-\mu)}{E}$$

$$= \frac{22.5 \times (2-0.3)}{100 \times 10^3} = \underline{3.825 \times 10^{-4}}$$

Change in diameter, $\delta_d = \varepsilon_c \times d = 3.825 \times 10^{-4} \times 45$

$= + \underline{0.0172 \text{ mm}}$ (increase)

B] Due to Pull of 3 kN (P=3kN):

Area of cross section of copper tube, $A_c = \pi \times d \times t$

$= \pi \times 45 \times 1.5 = 212.06 \text{ mm}^2$

Longitudinal strain, ε_L = direct stress/E = $\sigma/E = P/(A_c \times E)$

$= 3 \times 10^3/(212.06 \times 100 \times 10^3)$

$= \underline{1.415 \times 10^{-4}}$

Change in length, $\delta_L = \varepsilon_L \times L = 1.415 \times 10^{-4} \times 450 = \underline{+0.0637\text{mm}}$ (increase)

Lateral strain, $\varepsilon_{lat} = -\mu \times$ Longitudinal strain $= -\mu \times \varepsilon_L$

$= -\,0.3 \times 1.415 \times 10^{-4} = -4.245 \times 10^{-5}$

Change in diameter, $\delta_d = \varepsilon_{lat} \times d = -4.245 \times 10^{-5} \times 45$

$= \underline{-\,1.91 \times 10^{-3}\text{ mm}}$ (decrease)

C) Changes due to combined effects:

Change in length $= 0.0405 + 0.0637 = \underline{+\,0.1042\text{ mm}}$ (increase)

Change in diameter $= 0.01721 - 1.91 \times 10^{-3} = \underline{+\,0.0153\text{ mm}}$ (increase)

PROBLEM 3:

A cylindrical boiler is 800mm in diameter and 1m length. It is required to withstand a pressure of 100m of water. If the permissible tensile stress is $20 N/mm^2$, permissible shear stress is $8 N/mm^2$ and permissible change in diameter is 0.2mm, find the minimum thickness of the metal required. Take E = 200GPa, and $\mu = 0.3$.

SOLUTION:

Fluid pressure, $p = 100m$ of water $= 100 \times 9.81 \times 10^3 \; N/m^2$

$= 0.981 N/mm^2$.

1. Thickness from Hoop Stress consideration: (Hoop stress is critical than long. Stress)

$$\sigma_C = (p \times d)/(2 \times t)$$

$$20 = (0.981 \times 800)/(2 \times t)$$

$$t = \underline{19.62} \text{ mm}$$

2. <u>Thickness from Shear Stress consideration</u>:

$$\tau_{max} = \frac{(p \times d)}{(8 \times t)}$$

$$8 = \frac{(0.981 \times 800)}{(8 \times t)}$$

$$\therefore t = \underline{12.26mm.}$$

3. <u>Thickness from permissible change in diameter consideration</u> (δd=0.2mm):

$$\frac{\delta d}{d} = \frac{(p \times d)}{(4 \times t)} \times \frac{(2 - \mu)}{E}$$

$$\frac{0.2}{800} = \frac{(0.981 \times 800)}{(4 \times t)} \times \frac{(2 - 0.3)}{200 \times 10^3}$$

$$t = \underline{6.67mm}$$

Therefore, required thickness, t = <u>19.62</u> mm.

PROBLEM 4:

A cylindrical boiler has 450mm in internal diameter, 12mm thick and 0.9m long. It is initially filled with water at atmospheric pressure. Determine the pressure at which an additional water of 0.187 liters may be pumped into the cylinder by considering water to be incompressible. Take E = 200 GPa, and $\mu = 0.3$.

SOLUTION:

Additional volume of water, $\delta V = 0.187$ liters $= 0.187\times10^{-3}$ m^3

$= 187\times10^3$ mm^3

$$V = \frac{\pi}{4}\times 450^2 \times (0.9\times10^3) = 143.14\times10^6\ \text{mm}^3$$

$$\frac{dV}{V} = \frac{p\times d}{4\times t\times E}(5-4\times\mu)$$

$$\frac{187\times10^3}{143.14\times10^6} = \frac{p\times 450}{4\times12\times200\times10^3}(5-4\times0.33)$$

$\boxed{\text{Solving, } p=7.33\ \text{N/mm}^2}$